Question 1. Bootstrap Confidence Intervals for a Regression Slope

To investigate the uncertainty of a regression slope, generate $n = 200$ independent observations from

$$y_i = 1 + bx_i + \epsilon_i, \qquad x_i \sim U[-1, 1], \qquad \epsilon_i \sim \mathcal{N}(0, 1),$$

where x_i and ϵ_i are independent. Consider two cases: $b = 0$ and $b = 0.4$. Here, $\mathcal{N}(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2. Use a fixed random seed and report it.

(a) **Regression Estimation**

- Estimate the intercept and slope using ordinary least squares (OLS).
- Plot the observations and the fitted regression line for each value of b.

(b) **Bootstrap Resampling**

- Perform a pairs bootstrap with $B = 2000$ replicates. In each replicate, sample n indices with replacement and select the corresponding pairs (x_i, y_i) using the same indices.
- Refit the regression for each bootstrap sample and retain the estimated slope $\hat{b}^{*(j)}$, where $j = 1, \ldots, B$. Plot the bootstrap distribution of the estimated slope.

(c) **Confidence Intervals**

- Calculate the 95% percentile bootstrap confidence interval:

$$\left[q_{0.025}(\hat{b}^*), q_{0.975}(\hat{b}^*)\right],$$

  where q_p denotes the empirical p-quantile. Compare this interval with the conventional 95% OLS confidence interval:

$$\hat{b} \pm t_{0.975,n-2}\, \mathrm{SE}_{\mathrm{OLS}}(\hat{b}).$$

(d) **Interpretation and Resampling Assumptions**

- Explain why independently resampling the x_i and y_i values is inappropriate for this analysis.
- Does the ordinary bootstrap slope distribution directly represent the null distribution under $H_0 : b = 0$? Explain the distinction between estimating sampling uncertainty and constructing a null distribution.

Question 2. Regression Significance with AR(1) Errors

Let $n = 200$ and $x_t = -1 + 2(t-1)/(n-1)$ for $t = 1, \ldots, n$. Generate

$$y_t = 1 + 0.4x_t + \epsilon_t,$$
$$\epsilon_t = \phi\epsilon_{t-1} + u_t, \qquad u_t \sim \mathcal{N}(0, 1-\phi^2), \quad t = 2, \ldots, n,$$

with $\epsilon_1 \sim \mathcal{N}(0,1)$ independent of all innovations. Consider $\phi \in \{0, 0.5, 0.9\}$. The stationary residual variance is one for every ϕ. Treat ϕ and the noise variance as known. Test $H_0 : b = 0$ against $H_1 : b \neq 0$ at the 5% level.

(a) **Regression Estimation**

- Fit an OLS regression for each ϕ and report the estimated slope and conventional OLS standard error.

(b) **Significance Tests**

- Apply the conventional OLS t-test using $T = \hat{b}/\,\mathrm{SE}_{\mathrm{OLS}}(\hat{b})$ and a t distribution with $n-2$ degrees of freedom.
- For each ϕ, generate $B = 2000$ independent surrogate records under H_0 by replacing 0.4 with zero and retaining the same x_t, AR(1) parameters, and stationary initialization. Please visualize the histogram of regression slope, and locate the regression slope obtained by original OLS method.

Question 3. AR(1) versus AR(2) for an Oscillatory Signal

Generate $n = 400$ observations at integer times $t = 1, \ldots, 400$:

$$z_t = \sin\left(\frac{2\pi t}{20}\right) + v_t, \qquad v_t \sim \mathcal{N}(0, \sigma_v^2),$$

with independent measurement noise. Consider two noise levels, including $\sigma_v = 0$ and 0.3. Use the first 300 observations for training and reserve the rest for testing. Fit the models

$$\begin{aligned} \text{AR}(1): &\quad z_t = c + \phi_1 z_{t-1} + e_t, \\ \text{AR}(2): &\quad z_t = c + \phi_2 z_{t-2} + e_t. \end{aligned}$$

(a) **The Noise-Free Signal**

- Derive a second-order recurrence satisfied by the noise-free sinusoid. Explain how it motivates an AR(2) model.

(b) **Model Fitting**

- Fit both models by OLS with an intercept, using the same training response times $t = 3, \ldots, 300$. Report the coefficients and training residual root-mean-square error (RMSE) for each noise level.

(c) **Forecast Comparison**

- Produce a 40-step recursive forecast initialized at $t = 300$. Retain the training coefficients and use predicted values whenever subsequent lags extend beyond the training period. Plot both forecasts against the observations for $t = 301, \ldots, 340$ and report their RMSE over this interval.
- Which model better represents the oscillation for each noise level? Explain why training error alone is insufficient to select a model. Note that the noise-free sinusoid is a deterministic oscillation rather than a stationary AR(2) process driven by white-noise innovations.